Statements with Multiple Quantifiers

Quantifier order determines dependency.
Read the choices before manipulating the symbols.

1. Fix domains and predicates → 2. Preserve witness dependency → 3. Test truth or find failure → 4. Negate the whole condition → 5. Apply and check meaning

01 Meaning, domains and formalization

Predicate $P(x, y)$ has a truth value once its variables are assigned. A **multiply quantified statement** has at least two quantifiers. A **witness** makes an existential claim true.

Translate: choose domains; define argument order; identify "every/some/no"; preserve dependency; paraphrase back.

$\forall x \in D\ P(x) \equiv \forall x\,[x \in D \to P(x)]$
$\exists x \in D\ P(x) \equiv \exists x\,[x \in D \wedge P(x)]$

Example: every student knows a language.
$\forall s\,[\mathrm{Student}(s) \to \exists \ell\,(\mathrm{Language}(\ell) \wedge \mathrm{Knows}(s, \ell))]$
The $\exists$ means *at least one*, not exactly one. If $D = \varnothing$, the restricted $\forall$ statement is true; the restricted $\exists$ is false.

02 Two choice orders, two proof obligations

Dependent witness
$\forall x \in D\ \exists y \in E\ P(x, y)$
Any x is given first.
Then choose a suitable y.
y **may depend on** x**.**
Prove: for arbitrary x, give a working witness, often $y = f(x)$.
Disprove: find one x_0 for which *every* y fails: $\exists x \forall y\, \neg P(x, y)$.

Common witness
$\exists y \in E\ \forall x \in D\ P(x, y)$
Choose one y first.
Now any x may be given.
The same y **must work.**
Prove: give a fixed y_0 and verify it for arbitrary x.
Disprove: defeat *each* candidate y with an x: $\forall y \exists x\, \neg P(x, y)$.

Finite models: a universal step checks every candidate; an existential step needs one successful candidate. Continue recursively through the formula.

03 Equivalences, implication and countermodel

For fixed, independent domains, adjacent same-type quantifiers commute.
$\forall x \forall y\, P(x, y) \equiv \forall y \forall x\, P(x, y)$
$\exists x \exists y\, P(x, y) \equiv \exists y \exists x\, P(x, y)$
A common witness is sufficient, not necessary:
$\exists y \forall x\, P(x, y) \implies \forall x \exists y\, P(x, y)$

$D = E = \{1, 2\}$

$R(x,y)$	$y=1$	$y=2$
$x=1$	T	F
$x=2$	F	T

The converse fails.
$R = \{(1,1), (2,2)\}$. Each row has a true entry: choose $y = x$. No column is all true: no common y. Thus $\forall x \exists y\, R$ is true but $\exists y \forall x\, R$ is false.
Different quantifiers *can* give the same truth value in a particular model; there is no general equivalence.

04 English ↔ FOL: read the arguments

Here x, y range over people; $L(x, y)$ means "x loves y."

$\forall x \forall y\, L(x, y)$	Everyone loves everyone.
$\exists x \exists y\, L(x, y)$	Someone loves someone.
$\forall x \exists y\, L(x, y)$	Everyone loves at least one person.
$\exists x \forall y\, L(x, y)$	Someone loves everyone.
$\exists y \forall x\, L(x, y)$	Someone is loved by everyone.
$\forall y \exists x\, L(x, y)$	Everyone is loved by someone.

Negated predicate: preserve who acts on whom.

$\exists x \exists y\, \neg L(x, y)$	At least one ordered pair has no love relation.
$\forall x \forall y\, \neg L(x, y)$	Nobody loves anyone.
$\exists x \forall y\, \neg L(x, y)$	There is a person who loves nobody.
$\forall x \exists y\, \neg L(y, x)$	Everyone has someone who does not love them.

$\mathrm{LovedBy}(y, x) \equiv L(x, y)$. For people and movies, use separate domains: $\exists m \in \mathrm{Movies}\, \forall p \in \mathrm{People}\, L(p, m)$ means one movie is loved by everyone.

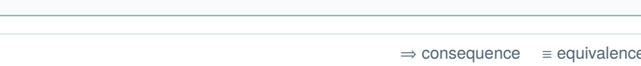
05 Numerical witnesses and boundary cases

Same predicate; different order (domain $\mathbb{Z}$)
$\forall x \exists y\,(x + y = 0)$: **TRUE.** Choose $y = -x$.
$\exists y \forall x\,(x + y = 0)$: **FALSE.** For any proposed y, choose $x = 1 - y$; then $x + y = 1 \neq 0$.

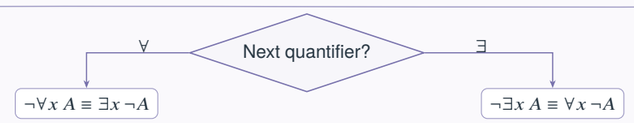
Reciprocal: b is a reciprocal of a exactly when $ab = 1$.
$\forall u \in \mathbb{R} \setminus \{0\}\ \exists v \in \mathbb{R}\,(uv = 1)$ **TRUE.**
Given $u \neq 0$, choose $v = 1/u$. The restriction is essential.
$\exists c \in \mathbb{R}\ \forall d \in \mathbb{R}\,(cd \neq 1)$ **TRUE.**
Witness $c = 0$: for all d, $0d = 0 \neq 1$.

Smallest positive integer
$\exists m \in \mathbb{Z}^+\ \forall n \in \mathbb{Z}^+\,(m \leq n)$ **TRUE**: choose $m = 1$.

No smallest positive real number
$\forall x \in \mathbb{R}^+\ \exists y \in \mathbb{R}^+\,(y < x)$ **TRUE.**
Choose $y = x/2$: $0 < y < x$. Zero is *not* positive.



06 Negation is an inward-moving procedure



Repeat for every quantifier; keep variable order and domains.
$\neg(Q_1 x_1 \cdots Q_n x_n\, P) \equiv \bar{Q}_1 x_1 \cdots \bar{Q}_n x_n\, \neg P$
Here $\bar{\forall} = \exists$ and $\bar{\exists} = \forall$. Then negate all remaining connectives (see 09).

$\neg\forall x \exists y\, P \equiv \exists x \forall y\, \neg P$
$\neg\forall x \forall y\, P \equiv \exists x \exists y\, \neg P$
$\neg\exists y \forall x\, P \equiv \forall y \exists x\, \neg P$
$\neg\exists x \exists y\, P \equiv \forall x \forall y\, \neg P$

Three choices: $\neg\forall x \exists y \forall z\, P \equiv \exists x \forall y \exists z\, \neg P$.
Not everyone loves someone $\equiv \exists x \forall y\, \neg L(x, y)$:
some person loves nobody, not just "does not love everyone."
Nobody loves everyone $\equiv \forall x \exists y\, \neg L(x, y)$:
each person fails to love at least one person.

07 Cafeteria: nested choices in a finite model

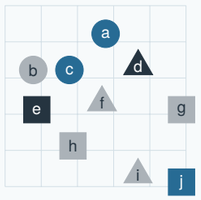
s: students; z: stations; i: available items. $\mathrm{Ch}(s, i)$: s chose i. $\mathrm{In}(i, z)$: i is in station z.

Student	Salads	Mains	Desserts	Drinks
Uta	green salad	spaghetti	pie	milk
Tim	fruit salad	fish	pie, cake	milk, coffee
Yuen	none	spaghetti, fish	pie	soda

A. An item chosen by every student
$\exists i \forall s\, \mathrm{Ch}(s, i)$ **TRUE**: the common witness is pie.
B. A student who chose every available item
$\exists s \forall i\, \mathrm{Ch}(s, i)$ **FALSE**: none chose all nine items.
C. Some student chose from every station
$\exists s \forall z \exists i\,[\mathrm{In}(i, z) \wedge \mathrm{Ch}(s, i)]$ **TRUE.**
Choose Uta or Tim; a suitable item may depend on the station.
D. Every student chose from every station
$\forall s \forall z \exists i\,[\mathrm{In}(i, z) \wedge \mathrm{Ch}(s, i)]$ **FALSE.**
Failure: choose Yuen and the salad station; no item works.

Fix s = Yuen → Fix z = salads → No witness i

08 Tarski world: geometry makes truth visible



Domain: the ten objects $a, \ldots, j$.
$C(x)$: circle; $S(x)$: square; $T(x)$: triangle.
$H(x, y)$: same color.
$\mathrm{Above}(x, y)$: higher row.
$\mathrm{RightOf}(x, y)$: larger column.
Colors: blue, gray, black. Relations are strict: an object is not above/right of itself.

$\forall x\,[C(x) \to \mathrm{Above}(x, f)]$ **TRUE**: all circles are above f.
$\exists x\,[S(x) \wedge \mathrm{Black}(x)]$ **TRUE**: witness e.
$\forall x\,[C(x) \to \exists y\,(S(y) \wedge H(x, y))]$ **TRUE.**
Witnesses: for a, c, use j; for b, use g or h.
$\exists x\,[S(x) \wedge \forall y\,(T(y) \to \mathrm{RightOf}(x, y))]$ **TRUE.**
A common square to the right of all triangles is g (or j).

Contrast: does every square have a same-color circle?
$\forall x\,[S(x) \to \exists y\,(C(y) \wedge H(x, y))]$ **FALSE.**
The black square e is a counterexample: there is no black circle.

09 Negating restrictions and compound conditions

Connective rules (apply inside the quantifiers)
$\neg(A \to B) \equiv A \wedge \neg B$
$\neg(A \wedge B) \equiv \neg A \vee \neg B$ $\quad \neg(A \vee B) \equiv \neg A \wedge \neg B$

Worked procedure (C, S, H as defined in 08)
$F = \forall x\,[C(x) \to \exists y\,(S(y) \wedge H(x, y))]$
$\neg F \equiv \exists x \neg[C(x) \to \exists y\,(S(y) \wedge H(x, y))]$
Flip $\forall$; then negate the implication.
$\equiv \exists x\,[C(x) \wedge \neg\exists y\,(S(y) \wedge H(x, y))]$
Flip $\exists$; then apply De Morgan's law.
$\equiv \exists x\,[C(x) \wedge \forall y\,(\neg S(y) \vee \neg H(x, y))]$
Meaning: a circle has no square of the same color. This is false in the displayed world; F is true.

Restriction traps
$\forall x\,[\mathrm{Student}(x) \wedge \mathrm{Passed}(x)]$ also asserts that *every object is a student*. It does not merely say all students passed.
Some student passed: $\exists x\,[\mathrm{Student}(x) \wedge \mathrm{Passed}(x)]$.
An existential restriction needs $\wedge$, not $\to$: a non-student would make $\mathrm{Student}(x) \to \mathrm{Passed}(x)$ true vacuously.

10 Computing: policies, constraints and rules

Access control $A(u, s)$: user u can access server s.
$\forall u \exists s\, A(u, s)$: every user can access some server.
$\exists s \forall u\, A(u, s)$: a common server is accessible by all.
The second implies the first; the converse need not hold.
$\neg\forall u \exists s\, A(u, s) \equiv \exists u \forall s\, \neg A(u, s)$:
there is a user who cannot access *any* server.

Database constraints (indicated domains are restricted)
$\forall o \in \mathrm{Orders}\ \exists c \in \mathrm{Customers}\ \mathrm{HasCustomer}(o, c)$.
A customer may vary by order. In contrast,
$\exists a \in \mathrm{Admins}\ \forall t \in \mathrm{Tables}\ \mathrm{Access}(a, t)$
requires a single administrator who can access all tables.

Logic programming: Prolog uses facts, rules, search and inference to answer queries. Formal rules specify what must hold; a query asks what follows from the knowledge base.

11 AI: from a knowledge base to a consequence

Fictional model: W=West; N=Nono; A=America. $\mathrm{Sells}(x, y, z)$: seller x, item y, buyer z.
$\forall x \forall y \forall z\,[\mathrm{American}(x) \wedge \mathrm{Weapon}(y) \wedge \mathrm{Sells}(x, y, z)$
$\wedge \mathrm{Hostile}(z) \to \mathrm{Criminal}(x)]$
$\exists m\,[\mathrm{Owns}(N, m) \wedge \mathrm{Missile}(m)]$
$\forall m\,[(\mathrm{Missile}(m) \wedge \mathrm{Owns}(N, m)) \to \mathrm{Sells}(W, m, N)]$
$\forall m\,[\mathrm{Missile}(m) \to \mathrm{Weapon}(m)]$
$\forall z\,[\mathrm{Enemy}(z, A) \to \mathrm{Hostile}(z)]$
$\mathrm{American}(W) \wedge \mathrm{Enemy}(N, A)$

Inference: choose a missile witness m_0. The rules give $\mathrm{Weapon}(m_0)$, $\mathrm{Sells}(W, m_0, N)$ and $\mathrm{Hostile}(N)$. Together with $\mathrm{American}(W)$, all conditions of the first rule hold.
Therefore $\mathrm{Criminal}(W)$ by instantiation and modus ponens.

All premises → Instantiate m_0 → $\mathrm{Criminal}(W)$

12 Transfer and final precision checks

Limits encode alternating choices ($a_n \in \mathbb{R}$)
$\lim_{n\to\infty} a_n = L$ means
$\forall \varepsilon > 0\ \exists N \in \mathbb{Z}^+\ \forall n \in \mathbb{Z}^+\,[n > N \to |a_n - L| < \varepsilon]$.
N may depend on ε, but must work for all later n.
For $a_n = 1/n$, $L = 0$: choose $N > 1/\varepsilon$; then $n > N$ gives $0 < 1/n < \varepsilon$.

"Someone else" needs inequality inside the scope:
$\forall x \exists y\,[x \neq y \wedge L(x, y)]$. Self-love is still allowed; to forbid it, also require $\forall x\, \neg L(x, x)$.
Using $x \neq y \to L(x, y)$ instead lets $y = x$ satisfy the condition vacuously; it does not require loving anyone else.

Remember: $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$, not $\forall x\, \neg P(x)$. A witness need not be unique or different for each input.
Orders can agree: for $P(x, y) : (x + y = x)$ over $\mathbb{Z}$, both $\forall x \exists y\, P$ and $\exists y \forall x\, P$ are true: $y = 0$ works for every x.